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The Performance of ChatGPT-4 and Gemini Ultra 1.0 for Quality Assurance Review in Emergency Medical Services Chest Pain Calls

Graham Brant-Zawadzki[a,b], Brent Klapthor[a], Chris Ryba[a,c], Drew C. Youngquist[a], Brooke Burton[b], Helen Palatinus[a], and Scott T. Youngquist[a,c]

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ABSTRACT

Objectives: This study assesses the feasibility, inter-rater reliability, and accuracy of using OpenAI's ChatGPT-4 and Google's Gemini Ultra large language models (LLMs), for Emergency Medical Services (EMS) quality assurance. The implementation of these LLMs for EMS quality assurance has the potential to significantly reduce the workload on medical directors and quality assurance staff by automating aspects of the processing and review of patient care reports. This offers the potential for more efficient and accurate identification of areas requiring improvement, thereby potentially enhancing patient care outcomes.

Methods: Two expert human reviewers, ChatGPT GPT-4, and Gemini Ultra assessed and rated 150 consecutively sampled and anonymized prehospital records from 2 large urban EMS agencies for adherence to 2020 National Association of State EMS metrics for cardiac care. We evaluated the accuracy of scoring, inter-rater reliability, and review efficiency. The inter-rater reliability for the dichotomous outcome of each EMS metric was measured using the kappa statistic.

Results: Human reviewers showed high interrater reliability, with 91.2% agreement and a kappa coefficient 0.782 (0.654-0.910). ChatGPT-4 achieved substantial agreement with human reviewers in EKG documentation and aspirin administration (76.2% agreement, kappa coefficient 0.401 (0.334-0.468), but performance varied across other metrics. Gemini Ultra's evaluation was discontinued due to poor performance. No significant differences were observed in median review times: 01:28 min (IQR 1:12 – 1:51 min) per human chart review, 01:24 min (IQR 01:09 – 01:53 min) per ChatGPT-4 chart review ($p = 0.46$), and 01:50 min (IQR 01:10-03:34 min) per Gemini Ultra review ($p = 0.06$).

Conclusions: Large language models demonstrate potential in supporting quality assurance by effectively and objectively extracting data elements. However, their accuracy in interpreting non-standardized and time-sensitive details remains inferior to human evaluators. Our findings suggest that current LLMs may best offer supplemental support to the human review processes, but their current value remains limited. Enhancements in LLM training and integration are recommended for improved and more reliable performance in the quality assurance processes.

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Introduction

Quality assurance (QA) in Emergency Medical Services (EMS) is crucial to ensure the effectiveness and safety of patient care. Traditionally, QA in EMS involves human professionals who review patient care reports against established clinical guidelines. However, manual review by trained clinicians is labor intensive and subject to human variability. Large language models (LLMs) are specific types of artificial intelligence tools trained on vast amounts of data to perform various natural language processing tasks. Recent advances in the development of LLMs that can analyze and process complex data raise the possibility that artificial intelligence (AI) may be able to serve as an aid in the QA process. ChatGPT, a LLM released by OpenAI (San Francisco, CA) and Google Gemini Ultra 1.0 released by Alphabet Inc (Mountain View, CA), use natural language processing driven by artificial intelligence to interact with users, engage in conversations, and provide human-like responses (1). The most recent version of ChatGPT at the time of this writing, GPT-4, was released in March 2023 and has shown promise in analyzing and interpreting complex data. For example, ChatGPT-4 has demonstrated the ability to well-exceed passing scores on the United States (U.S.) Medical Licensing Examinations and achieve near passing scores and the ability to offer relevant explanations to scenario-based questions on the American Heart Association Basic Life Support and Advanced Cardiovascular Life Support exams (2, 3). Gemini Ultra 1.0 was released on 2/8/2024 as a competitor.

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In the pre-ChatGPT era, a 2020 pilot study found favorable results in using a variety of LLM techniques to review prehospital records in assessing the appropriateness of prehospital care provided for trauma patients (4). Other studies have indicated that LLMs are capable of reviewing patient records for disease phenotyping and have successfully reduced both false positives and false negatives in disease diagnosis (5, 6). Given these prior successes, we hypothesized that the ChatGPT-4 LLM would be capable of automated, objective, and efficient quality review of EMS patient care reports. This study aims to explore the feasibility and accuracy of using ChatGPT-4 as an EMS QA professional, comparing its performance against human EMS professionals in reviewing patient care reports for the complaint of chest pain.

Methods

Study Design and Setting

This study is a comparative agreement analysis between the ability of human reviewers and two chat-based LLM systems to review medical records accurately and efficiently for documentation of key quality metrics for cases of suspected cardiac chest pain managed by 2 large urban EMS agencies operating in a large urban setting. This study was deemed minimal risk and performed under IRB approval from the University of Utah IRB (IRB_00138043). The study setting is a large urban corridor with a population of roughly 1.2 million people. Prehospital care in the county is provided by 11 ground-based EMS agencies responding to urban, suburban, and rural settings. Two agencies provide care to the most densely populated regions, responding to the majority of over 80,000 Public Safety Access Point (9-1-1) calls per year. Both agencies perform continuous quality assurance *via* manual review of electronic patient care reports evaluating adherence to local protocols and quality metrics as recommended by Version 3.0 of the National Association of State EMS Officials 2022 National Model EMS Clinical Guideline (7).

Selection of Participants

We collected a consecutive sample of the first 150 electronic patient care records (EPCR) from two large EMS urban agencies with samples approximately proportionate to each fire department's overall run volume and representing roughly 2% of each agency's total call volume for over the same time period. We chose chest pain as our call type of interest given its importance in the systems-based care of ST-Elevation Myocardial Infarction (STEMI) (8). We screened consecutive records based upon a documented primary impression of STEMI ($N = 30$) or non-traumatic chest pain in patients aged 35 or older between January 1, 2023 to February 1, 2024. Independent reviewers then read the individual candidate EPCR and assessed each for inclusion based on the above criteria. The initial screening process included complaints of non-traumatic chest pain, but following the review process, those EPCRs not related to non-traumatic chest pain were excluded. We further evaluated candidate EPCRs for inclusion in the study by reading the Narrative section of the reports and excluded reports that provided an impression other than non-traumatic chest pain.

We downloaded the EPCR in PDF format from each agency's respective electronic health record system (ImageTrend, Inc, Lakeville, MN and ESO Solutions, Austin, TX). To ensure privacy, all patient-identifying information was anonymized following guidance from the U.S. Department of Health and Human Services document entitled Guidance Regarding Methods for De-identification of Protected Health Information in Accordance with the Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule (9). To do this, we removed identifiers using a customized downloadable report prior to PDF export for each EPCR and/or by redacting any other potential patient or clinician identifiers found in the reports using permanent markup tools in Adobe Acrobat Pro (Adobe, Inc, San Jose, CA) or Apple Preview (Version 11.0, Apple Corp, Cupertino, CA). Anonymization was performed prior to uploading cases to the web-based LLMs for interrogation.

Interventions

We developed an online Microsoft Forms survey instrument for data entry (Microsoft365 Suite, Microsoft Corp, Redmond, WA) based on key quality metrics for the complaint of "Chest Pain/Acute Coronary Syndrome (ACS)/ST-segment Elevation Myocardial Infarction" found on page 33 of Version 3.0 of the U.S. Model National Model Clinical EMS Clinical Guidelines developed by the National Association of State EMS Officials (7). The survey instrument is available in the Supplemental Appendix.

In addition to quality metrics, we documented basic demographic information for each case. We used logical gating for some questions. For example, if the reviewer answered that the performance of an ECG was not documented, the question regarding whether the timing of the ECG was documented would be logically skipped. Most questions required binary, yes/no responses or were yes/no/not applicable with a few free-form text boxes for the entry of documented times.

Two board-certified EMS medical directors (GBZ and CR) reviewed the selected EPCRs for the respective agencies for which they provide regular medical direction and quality assessment. Each performing their typical QA assessment and then entered data into the survey instrument. To assess inter-rater reliability, each reviewer subsequently cross-reviewed 10 of the cases initially reviewed by the other (20 cross-reviewed, 13% of sample). The human reviewers were not blinded to the study hypothesis but were blinded to the results of each other's reviews. Because cross-review of the 20 cases provided two criterion standards for these cases, a third reviewer (SY) adjudicated any discrepancies and eliminated duplicate responses before comparing these cases to LLM responses.

Beginning with the key quality metrics for "Chest Pain/Acute Coronary Syndrome (ACS)/ST-segment Elevation Myocardial Infarction" in Version 3.0 of the U.S. Model National Model Clinical EMS Clinical Guidelines, an

unstructured process of iterative prompt testing was performed by two of the investigators (BK and SY). Neither investigator had any prior expertise in LLM training or prompt design. This included repeated informal real-time testing of anonymized charts against human review. At times the LLM's errors were noted back to it in prompting, asking for feedback on how to improve upon the prompt. The prompting included in Figure 1 was settled on when repeated testing of changes failed to note improvements in the LLM review. Once this was done, the same EPCRs were uploaded, one by one, to either ChatGPT-4 or Gemini Ultra 1.0 between March 1st, 2024 and March 12th, 2024. framed by the prompt. Two of the investigators (DCY and BK) uploaded the records, prompted the LLM, and entered responses into the survey instrument before moving on to the next question. These reviewers were not blinded to the study hypothesis but were blinded to the responses of human reviewers.

After the performance of 47 reviews using Gemini Ultra 1.0 we abandoned the effort due to poor performance. While ChatGPT-4 allowed for direct upload of PDF documents into the interface, Gemini Ultra 1.0 required uploading of documents to Google One Drive. Gemini Ultra 1.0 repeatedly "hallucinated" reviews where it reported to have found the correct file in Google One Drive but in fact had not found the file and fabricated answers (10). Repeated attempts to correct this error were unsuccessful.

Measurements

For the purposes of this study, the medical director review was considered the criterion standard against which the performance of the LLM would be measured. When querying the LLMs, we presented all survey questions simultaneously within the prompt and did not perform the type of logic gating employed for human review. This allowed for an error in one question to create cascading errors in other questions. For example, if the LLM answered incorrectly that an ECG was not performed, it was still asked if the timing of the performance of an ECG was documented. We chose this approach based on the known ability of LLMs to "hallucinate" answers and to implicitly test the LLMs ability to reason appropriately (10).

We estimated the time taken to perform each review by both human and the LLM system by using the recorded times from survey opening to completion in Microsoft Forms. Reviewers were informed that reviews were timed and not to step away from an open survey once started to avoid abnormally long review times.

Outcomes

The primary outcome of interest was the Cohen's kappa statistic for overall agreement between the LLM and human review. Secondary outcomes include 1) the kappa statistic and percent agreement of each key quality metric and 2) comparisons of the distributions of the time required for case review between reviewers (human, ChatGPT-4, Gemini).

Analysis

An a priori sample size estimate was not performed given the paucity of data available on this topic to inform power calculations.

[upload redacted patient care report]

I am uploading a patient care report from EMS. I want you to act as a quality assurance officer and answer my questions regarding the quality of care. Ready?

[enter]

I would like you to answer the following questions:

1. By reviewing the PDF documentation thoroughly with particular attention to the narrative text, can you determine the patient's time of symptom onset?
2. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or interventions under the subheadings of "Activities" or "Flow Chart," can you determine if the performance of a 12-lead EKG was documented?
3. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or timing of interventions listed under the subheadings of "Activities" or "Flow Chart," can you determine the exact time the 12-lead EKG was performed? If yes, tell me what it is. If not, tell me why it cannot be determined.
4. What is the time elapsed between patient contact by EMS to the time of 12-lead EKG acquisition? This will require you to find the timestamp for EMS arrival "At Scene" or "On Scene" and calculate the time interval between this time and the time when the EKG was performed which you answered in the prior question. Calculate this time interval for time to 12-lead acquisition in your answer to this question.
5. By reviewing the PDF documentation thoroughly, can you determine if the patient received aspirin? For example, aspirin may have been taken prior to EMS Arrival, given by EMS, or there may be a documented contraindication to aspirin such as an allergy.
6. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or timing of interventions listed under the subheadings of "Activities" or "Flow Chart", can you determine the exact time of Aspirin administration? If yes, tell me what it is. If not, tell me why it cannot be determined.
7. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or timing of interventions listed under the subheadings of "Activities" or "Flow Chart", determine if there was a STEMI diagnosed? If so what was the time of STEMI notification to the receiving hospital? The answer to this question will be "not applicable" if no STEMI was diagnosed.

[enter answers into Google Form]

Ok, we are finished with that prior PDF. Here is the pdf of the next patient care report from EMS. Are you ready to answer my questions regarding the quality of care?
[insert next PDF] [enter]
[then just give it the same prompt of 7 questions & repeat]

I would like you to answer the following questions:

1. By reviewing the PDF documentation thoroughly with particular attention to the narrative text, can you determine the patient's time of symptom onset?
2. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or interventions under the subheadings of "Activities" or "Flow Chart," can you determine if the performance of a 12-lead EKG was documented?
3. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or timing of interventions listed under the subheadings of "Activities" or "Flow Chart," can you determine the exact time the 12-lead EKG was performed? If yes, tell me what it is. If not, tell me why it cannot be determined.
4. What is the time elapsed between patient contact by EMS to the time of 12-lead EKG acquisition? This will require you to find the timestamp for EMS arrival "At Scene" or "On Scene" and calculate the time interval between this time and the time when the EKG was performed which you answered in the prior question. Calculate this time interval for time to 12-lead acquisition in your answer to this question.
5. By reviewing the PDF documentation thoroughly, can you determine if the patient received aspirin? For example, aspirin may have been taken prior to EMS Arrival, given by EMS, or there may be a documented contraindication to aspirin such as an allergy.
6. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or timing of interventions listed under the subheadings of "Activities" or "Flow Chart", can you determine the exact time of Aspirin administration? If yes, tell me what it is. If not, tell me why it cannot be determined.
7. By reviewing the PDF documentation thoroughly with particular attention to the narrative text or timing of interventions listed under the subheadings of "Activities" or "Flow Chart", determine if there was a STEMI diagnosed? If so what was the time of STEMI notification to the receiving hospital? The answer to this question will be "not applicable" if no STEMI was diagnosed.

[enter answers into Google Form]

Figure 1. The LLM prompt used by investigators in this study.

We recorded data on an Excel spreadsheet (Excel for Mac Version 16.83, Microsoft Corp, Redmond, WA) and imported the data into Stata statistical software (Stata/BE 18.0 for Mac, StataCorp, College Station, TX) for analysis.

We measured inter-rater reliability for the dichotomous outcome of each EMS metric using the kappa statistic (11). The kappa coefficient ranges from 0 (amount of agreement expected by random chance) to 1 (perfect agreement). Scores between 0.61-0.80 indicated substantial agreement while those between 0.81-1.00 reflected near perfect agreement (12). We reported the results using a 95% confidence interval. We calculated both percent agreement and expected agreement between the LLM and human reviewers. Additionally, we assessed the LLM's validity compared to human review by determining the sensitivity and specificity. Finally, we used the Mann Whitney U test to compare the distribution of time to complete the survey instrument between groups. We considered p values >0.05 to be not statistically significant.

Results

Characteristics of Study Subjects

The analysis encompassed a review of 150 medical records (Table 1).

Human Review

The two EMS directors demonstrated almost perfect agreement on all EMS metrics, except for the documentation of time of onset of symptoms [70% agreement; kappa coefficient 0.400 (95% CI, -0.002-0.802)] and time of administration of aspirin [80% agreement; kappa coefficient 0.444 (95% CI, -0.080-0.969)] (Table 2).

ChatGPT Review

Inter-rater reliability varied between ChatGPT-4 and the EMS director reviewers. While there was substantial agreement between the LLM and human reviewers for the metrics of EKG documentation [94.7% agreement; kappa coefficient 0.704 (95% CI, 0.510-0.897)] and administration of aspirin [93.3%; 0.792 (95% CI, 0.669-09.15)], there was variable performance in the other metrics (Table 3). The LLM had moderate agreement with humans in the documentation of symptom onset [76% agreement; kappa coefficient 0.426 (95% CI, 0.280-0.572)]. However, it performed fair to poor in assessing time-related elements: time to performance of EKG documentation [68.4% agreement; kappa coefficient 0.323 (95% CI, 0.184-0.463)], time of aspirin administration documentation [67.2% agreement, kappa coefficient 0.146 (95% CI, −0.031-0.324)], time to documentation of STEMI notification [64.7% agreement; kappa coefficient 0.164 (95% CI, −0.109-0.437)]. ChatGPT-4 had the worst inter-rater reliability in assessing the documentation of time to acquisition of EKG [53.2% agreement; 0.016 (95% CI, -0.156-0.188)].

Gemini Review

Forty-seven medical records were reviewed by Google Gemini Ultra 1.0 prior to the decision to abandon efforts. The LLM had moderate agreement with human reviewers in assessing whether aspirin administration was documented [80.9% agreement; kappa coefficient 0.557 (95% CI, 0.308-0.806)]. It performed worse than that which would have been expected by random chance in the assessment of time of EKG documentation [72.7% agreement; kappa coefficient −0.015 (95% CI, -0.298-0.268)]. Google Gemini Ultra 1.0 had slight to fair agreement with human reviewers on all other metrics (Table 4).

Table 1. Demographic and clinical characteristics of reviewed chest pain EMS electronic patient care reports ($n = 150$).

Characteristic	
Age (SD)	64.7 (15)
Female Gender (%)	63 (42)
Race/Ethnicity	
American Indian or Alaskan Native (%)	1 (1)
Asian (%)	4 (2.7)
Black/African American (%)	4 (2.7)
Caucasian (%)	104 (69)
Hispanic/Latino (%)	10 (6.7)
Pacific Islander/Hawaiian (%)	1 (1)
Unknown (%)	26 (17)
Disposition	
Transported (%)	138 (92)
Released on Scene (%)	12 (8)
Time of Onset Documented (%)	95 (63)
Performance of a 12-Lead ECG Documented (%)	134 (89)
Time of 12-Lead ECG Documented, $n = 134$ (%)	107 (80)
12-Lead ECG Performed within 10 min, $n = 107$ (%)	79 (74)
Aspirin Administered or Contraindication Documented	117 (78)
Time of Aspirin Administration Documented, $n = 114$ (%)	95 (83)
Time of STEMI Documented	
Yes (%)	15 (10)
No (%)	33 (22)
Not Applicable/No STEMI Identified (%)	102 (68)
Time from first diagnostic 12-lead ECG to STEMI Notification (min), $n = 14$, (IQR)	5.5 (1.25-10.75)

Table 2. Inter-rater agreement between EMS director reviews for chest pain quality metrics.

Metric	Agreement, Number (%)	Expected Agreement, %	Kappa Coefficient (95% CI)	Sensitivity, %	Specificity, %
Was the time of onset of symptoms documented? ($n = 20$)	14 (70.0)	50.0	0.400 (-0.002-0.802)	70	70
Was the performance of a 12-lead EKG documented? ($n = 20$)	20 (100.0)	82.0	***	100	100
Was the time of performance of the 12-lead EKG documented? ($n = 18$)	18 (100.0)	65.4	***	100	100
Was the time of patient contact to the time of 12-lead EKG acquisition within 10 min? ($n = 14$)	13 (92.9)	62.2	.811 (0.460-1.000)	100	75
Did the patient receive aspirin? ($n = 20$)	20 (100.0)	68.0	***	100	100
Was the time of aspirin documented? ($n = 15$)	12 (80.0)	64.0	0.444 (-0.080-0.969)	90.9	50
Was the time of STEMI notification documented? ($n = 6$)	6 (100.0)	72.2	1.000 (1.000-1.000)	100	100

***In cases with perfect agreement among raters, the confidence interval calculation was limited due to the lack of variability within the data set.

Table 3. Inter-rater agreement between ChatGPT-4 and EMS medical directors for chest pain quality metrics.

Metric	Agreement, Number (%) $n = 150$	Expected Agreement, %	Kappa Coefficient (95% CI)	Sensitivity, %	Specificity, %
Was the time of onset of symptoms documented? ($n = 150$)	114 (76.0)	58.2	0.426 (0.280-0.572)	94.7	43.6
Was the performance of a 12-lead EKG documented? ($n = 150$)	142 (94.7)	82.0	0.704 (0.510-0.897)	97.8	68.8
Was the time of performance of the 12-lead EKG documented? ($n = 139$)	95 (68.4)	53.2	0.323 (0.184-0.463)	64.9	82.1
Was the time of patient contact to the time of 12-lead EKG acquisition within 10 min? ($n = 109$)	58 (53.2)	52.5	0.016 (-0.156-0.188)	55.6	46.4
Did the patient receive aspirin? ($n = 150$)	140 (93.3)	67.9	0.792 (0.669-09.15)	98.3	75.8
Was the time of aspirin documented? ($n = 125$)	84 (67.2)	61.6	0.146 (-0.031-0.324)	72.3	45.8
Was the time of STEMI notification documented? ($n = 51$)	33 (64.7)	57.8	0.164 (-0.109-0.437)	33.3	81.8

Overall Agreement

Among the three pairs of reviewers, there was the highest overall agreement between the EMS directors [91.2% agreement; kappa coefficient 0.782 (95% CI, 0.654-0.910)] followed by fair agreement between ChatGPT-4 and human reviewers [76.2% agreement; kappa coefficient 0.401 (95% CI 0.334-0.468)] (Table 5).

Accuracy

ChatGPT-4 and Google Gemini Ultra 1.0 possessed variable degrees of accuracy in the evaluation of EMS metrics (Tables 3 and 4). The LLMs had the highest sensitivity and specificity in assessing the documentation of aspirin administration (98.3% sensitivity, 75.8 specific for ChatGPT GPT and 80.0% sensitivity, 83.3% specific for Google Gemini Ultra 1.0) but to a lesser degree in all the other metrics.

Time to Review

There were no differences in the distribution of times to complete reviews between human review, median 01:28 min (IQR 1:12-01:51 min) and the GPT model review process, median time of 01:24 min (IQR 1:09-01:53 min) ($p = 0.46$). There was also no difference in the distribution of times between human review and Gemini Ultra 1.0, median 01:50 (IQR 01:10-03:34 min), $p = 0.06$.

Discussion

The utilization of LLMs in the QA process of EMS shows promise, particularly in standardized data element extraction. However, the current analysis suggests that LLMs, as represented by ChatGPT-4, although efficient, encounter limitations in discerning nuanced details, such as the exact times of procedures and notifications.

Table 4. Inter-rater agreement between Google Gemini Ultra 1.0 and EMS medical directors for chest pain quality metrics.

Metric	Agreement, Number (%)	Expected Agreement, %	Kappa Coefficient (95% CI)	Sensitivity, %	Specificity, %
Was the time of onset of symptoms documented? ($n = 47$)	33 (70.2)	55.1	0.337 (0.055-0.618)	77.4	56.3
Was the performance of a 12-lead EKG documented? ($n = 47$)	42 (89.4)	82.5	0.393 (-0.027-0.812)	97.6	33.3
Was the time of performance of the 12-lead EKG documented? ($n = 44$)	32 (72.7)	73.1	−0.015 (-0.298-0.268)	81.6	16.7
Was the time of patient contact to the time of 12-lead EKG acquisition within 10 min? ($n = 38$)	26 (68.4)	64.5	0.109 (-0.173-0.392)	70.6	50
Did the patient receive aspirin? ($n = 47$)	38 (80.9)	56.8	0.557 (0.308-0.806)	80.0	83.3
Was the time of aspirin documented? ($n = 37$)	25 (67.6)	63.0	0.123 (-0.216-0.461)	75.9	37.5
Was the time of STEMI notification documented? ($n = 14$)	7 (50.0)	39.8	0.169 (-0.073-0.412)	22.2	100

Table 5. Overall agreement among reviewers for all EMS chest pain quality metrics.

Rater	Agreement, Number (%) $n = 150$	Expected Agreement, %	Kappa Coefficient (95% CI)	Sensitivity, %	Specificity, %
EMS Director – EMS Director	103 (91.2)	59.4	0.782 (0.654-0.910)	95.0	81.8
ChatGPT GPT-4 – EMS Director	666 (76.2)	60.3	0.401 (0.334-0.468)	81.0	61.8
Google Gemini Ultra 1.0 – EMS Director	203 (74.1)	62.6	0.308 (0.183-0.432)	78.8	56.1

Human reviewers exhibited high consistency in non-time-related elements, which aligns with the focused training and expertise that human professionals bring to the review process. One thought behind these discrepancies is that some of the time sensitive actions have multiple elements to their documentation, e.g., administration of aspirin could have been performed and documented prior to EMS arrival in both a full or partial dose and may have completed the partial dose with the time of administration being either that initial dose or completed dose. Prior to arrival occurrences of 12 lead EKGs (by a clinic or other health care agency or administration of aspirin could have led to time values not easily interpreted by LLM. The LLM, while expedient, showed discrepancies in interpreting and verifying time-sensitive actions, crucial in cardiac-related emergencies where minutes can dictate patient outcomes. The differing levels of agreement on the documentation of times such as symptom onset, 12-lead EKG performance, and STEMI notification, highlight areas where LLMs may require further training or more advanced algorithms to match human accuracy.

The slight advantage of LLMs in processing speed must be weighed against these discrepancies. The efficiency gains of LLMs are notable, but they should not compromise the fidelity of data extraction, which is essential for patient care and outcome analysis. Therefore, while LLMs have a role in augmenting the QA process by handling more routine tasks, human oversight remains vital for ensuring the quality of patient care reports, especially in aspects requiring clinical judgment.

The results also emphasize the need for clear documentation practices, as variability in record completeness can influence the performance of both human and LLM reviewers. Consistency in documentation is key to enabling both LLMs and human professionals to perform at their best. LLMs offer a litany of other potential applications. Documentation, for example, ß may be improved in the future through natural language, processing LLM's that listen and document the encounter. Software such as Freed (https://www.getfreed.ai/, accessed 5/19/2024) listens to the encounter and perform medical documentation for physicians. Such software may soon be available in the prehospital setting, although encounters and uncontrolled setting for the significant background noise, and with an uncooperative patience, may make this technically challenging.

Limitations

The methodology of our study introduces several limitations that warrant consideration. The reviews were conducted in a single session, which, although efficient, may not accurately mirror the typically more distributed nature of reviews in real-world EMS quality assurance processes. This batched approach could have increased reviewer fatigue and error rates, potentially skewing the performance metrics in favor of LLM review. Though, ultimately both human and LLM reviews required human input into the survey instrument allowing for human data input error. Further, investigators had no formal expertise in the use and prompting of the LLM tools. LLM tools specifically designed and extensively trained by physicians and QA experts may perform better than the open-resource tools used here.

Additionally, the human reviewers, who were coauthors, medical directors associated with each agency, and intimately involved in all aspects of this study, were not blinded

to its hypotheses or objectives. This lack of blinding may have introduced bias, particularly in subjective assessments where "best-faith effort" was recorded, and when reviewers interpreted documentation that relied on vague or incomplete information from patients or bystanders.

Human and LLM reviewers inherently differ in their decision-making processes. Human reviewers can apply clinical experience to infer contextual nuances that may be absent in the explicit data inputs available to LLM models. Moreover, the training data and the specificity of the prompts used to guide the LLM's review process may not capture all nuances of real-world EMS documentation, potentially limiting the LLM's performance.

The LLM's natural language processing capabilities are also constrained by the quality of the text input. Factors such as poor syntax, inappropriate medical jargon, or unconventional and inconsistent abbreviations prevalent in EMS documentation could have impeded the LLM's ability to accurately assess the records.

Another limitation arises from the fact that, while extensive time was spent troubleshooting the LLM prompts, none of the authors possess formal training in the burgeoning field of artificial intelligence and LLM prompting. This lack of specialized expertise might have impacted the effectiveness and precision of the LLM prompts used in the study, potentially affecting the LLM's performance.

Furthermore, the reported review times for the LLMs likely significantly underestimate the potential time savings of a theoretical software capable of automatically reviewing and outputting QA metrics. The reported times included the duration required for individual human prompting of the LLM with questions and the uploading of the EPCR, as well as human input of results into our survey instrument. An automated system could potentially streamline these processes, thus enhancing the efficiency of LLM-driven reviews.

Finally, the Health Insurance Portability and Accountability Act (HIPAA), along with local guidelines, mandates strict rules for protecting patient information. To ensure patient privacy in our experiment, each EPCR was meticulously scrubbed of any potentially identifying information through a detailed, multistep process. This anonymization process was time-consuming and was not included in the review time because, in this study, we aimed solely to assess the capabilities of the LLMs in participating in the QA process. Consequently, the time required for anonymization significantly extended the overall time per chart review, which is impractical for real-world applications. HIPAA compliant LLMs are a requirement and must be developed for these models to be useful to quality professionals. Additionally, the LLMs utilized for this study were developed by third party organizations, raising concerns about who maintains usage rights to data incorporated during processing. Lastly, for real-world application, process for informing patients of how their data may be used, including the potential review of their medical data by LLMs, must be developed and put in place.

Future studies could mitigate these limitations by incorporating multiple review sessions to better simulate real-world conditions, using a broader and more varied dataset for training the LLM, and increasing the sample size with multiple blinded reviewers to minimize bias and enhance the reliability of findings. Additionally, exploring complex clinical decision-making tasks, a critical component of the QA process, could provide further insights into the comparative effectiveness of human versus LLM reviewers.

Conclusions

For LLMs to be integrated successfully, they should complement, not replace, the human element within the QA process of EMS. Based on our findings, we recommend integrating current-generation LLMs into the QA process as a preliminary review tool. These models can tag individual charts and identify larger trends for further analysis. Once flagged, human reviewers focus their efforts on more complex and nuanced evaluations. Ongoing training and calibration of these systems, informed by human reviewer feedback, will enhance the accuracy of individual LLMs over time. Greater collaboration between LLM developers and health care professionals will be essential to refine LLM capabilities and address current limitations on a larger scale. As LLM technology evolves, it is anticipated that it will become more adept at handling complex tasks, eventually leading to a more robust and efficient QA process.

The study's findings suggest an optimistic trajectory for LLM application in health care QA, contingent upon the continual improvement of LLMs' capabilities and an iterative process involving both technological and clinical expertise. Future research should focus on identifying the specific parameters that enhance the LLM's performance to ensure that LLM support becomes an indispensable asset in EMS QA processes.

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Decleration of Generative AI in Scientific Writing

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